

Boston Liveability Dashboard

Neighborhood-level socioeconomic analysis across 41 Boston zip codes

Power BI · Excel · US Census API · ACS 5-year estimates · 41 zip codes

OVERVIEW

Built a Power BI dashboard analyzing liveability across 41 Boston-area zip codes using US Census / ACS socioeconomic data collected via APIs. Processed raw data through a min-max normalization pipeline in Excel and produced an interactive multi-metric dashboard to surface neighborhood-level insights.

DATA AND METHODOLOGY

Source: US Census Bureau / ACS 5-year estimates. One row per zip code, collected via Census API, mapped to 41 Boston-area neighborhoods.

Dimensions analyzed:

- Median Household Income
- Education Attainment (% bachelor's degree or higher)
- Unemployment Rate
- Median Home Value
- Population Density (persons per sq. mile)

Normalization: Each dimension was min-max normalized to a 0-1 scale. Density and unemployment were inverted (lower = higher score) since both reduce liveability.

Composite Score Weights:

- Income - 30%
- Unemployment - 25%
- Education - 20%
- Home Value - 15%
- Population Density - 10%

KEY FINDINGS

- Cambridge / Downtown corridor consistently leads across income, education, and unemployment metrics.
- Roxbury, Mattapan, and Dorchester score lowest on the composite - driven primarily by income and unemployment gaps.
- High home values do not always signal high liveability: South End carries a \$946K median home value alongside 9.1% unemployment and below-median income.

- Fenway/Back Bay shows anomalously high unemployment (11%) despite affluent surroundings - likely driven by its large student population.
- North End / Waterfront is the density outlier at 64,500 persons/sq mile - scored down on density despite strong income and education metrics.

DASHBOARD FEATURES

- Choropleth maps (TomTom basemap) showing composite liveability score and individual metrics by zip code.
- Ranked bar charts for all 5 dimensions across all 41 neighborhoods.
- Interactive variable selector: toggle between Education, Income, Home Value, Density, and Unemployment on a single map.
- Outlier callouts for neighborhoods where metrics diverge from expectations.

TOOLS AND STACK

Power BI · Microsoft Excel (normalization pipeline) · US Census API · ACS 5-year estimates · TomTom Maps